ENSAI
Supervised Learning

Problem 1. k-Nearest Neighbors and Feature Scaling

Consider the following training dataset with two numerical features and a binary label.

Observation	X_1	X_2	Y
A	1	8	0
B	3	3	1
C	5	5	1
D	7	2	0
E	8	7	0

We want to classify a new observation $x = (4, 4)$.

1. Calculate the Euclidean distance from x to each training observation.
2. Determine the predictions of the 1-NN and 3-NN classifiers.
3. Now replace X_1 by $X_1' = 10X_1$, for both the training and test observations. Recalculate the distances and predictions.
4. Explain why feature scaling can influence k-NN predictions.

Problem 2. Naive Bayes and Posterior Probabilities

A medical screening system predicts whether a patient has a certain condition ($Y = 1$) using three binary symptoms X_1, X_2, X_3.

The estimated probabilities are:

Class	Prior	$P(X_1 = 1 \mid Y)$	$P(X_2 = 1 \mid Y)$	$P(X_3 = 1 \mid Y)$
$Y = 1$	0.3	0.8	0.7	0.6
$Y = 0$	0.7	0.2	0.4	0.3

Assume conditional independence of the symptoms given the class.

A patient has symptoms $X_1 = 1$, $X_2 = 0$, and $X_3 = 1$.

1. Calculate
$$P(X_1 = 1, X_2 = 0, X_3 = 1 \mid Y = 1).$$
2. Calculate the corresponding probability given $Y = 0$.
3. Use Bayes' formula to compute
$$P(Y = 1 \mid X_1 = 1, X_2 = 0, X_3 = 1).$$

4. What class would the Naive Bayes classifier predict?
5. Explain why the denominator in Bayes' formula can be ignored when choosing the most likely class, but not when computing the posterior probability.

Problem 3. Decision Trees and Gini Impurity

A classification tree is trained on a dataset containing three classes: 0, 1, and 2.

A node contains 20 observations: 8 from class 0, 7 from class 1, and 5 from class 2.

Two candidate splits produce the following child nodes:

Split	Child	Class 0	Class 1	Class 2
A	Left	6	2	2
A	Right	2	5	3
B	Left	5	5	0
B	Right	3	2	5

Recall that the Gini impurity is

$$G(t) = 1 - \sum_{j=0}^{2} p_j(t)^2,$$

where $p_j(t)$ is the proportion of observations from class j in node t.

1. Calculate the Gini impurity of the parent node.
2. Calculate the Gini impurity of each child node.
3. Calculate the weighted Gini impurity after each candidate split.
4. Which split should CART select? Justify your answer numerically.

Problem 4. Cost-Sensitive Classification

A company uses a classifier to detect potentially fraudulent transactions.

For a particular transaction, the estimated probability of fraud is

$$p(x) = P(Y = 1 \mid X = x) = 0.15.$$

The costs of classification errors are:

- False negative (fraud classified as legitimate): 40 units.
- False positive (legitimate transaction classified as fraud): 6 units.
- Correct classification: zero cost.

1. Calculate the expected cost of predicting class 0 (legitimate).
2. Calculate the expected cost of predicting class 1 (fraud).
3. Which prediction minimizes the expected cost?

4. Derive the optimal probability threshold t for predicting fraud as a function of the false-positive and false-negative costs.

5. How would the decision change if the false-negative cost were reduced to 20 units?

Problem 5. Regression Trees and Squared Error

A regression tree is trained using the following observations:

X	Y
1	2
2	4
3	3
4	7
5	8
6	6

We consider two candidate splits:

- Split A: $X \leq 3$.
- Split B: $X \leq 4$.

For a node t, its impurity is the average squared deviation from its mean:

$$I(t) = \frac{1}{N_t} \sum_{i \in t} (Y_i - \bar{Y}_t)^2.$$

1. Calculate the prediction of the regression tree before splitting.

2. Calculate the mean response in each child node under Split A.

3. Calculate the corresponding means under Split B.

4. Compute the weighted impurity after each split.

5. Which split provides the greater reduction in squared error?

6. What would be the prediction for $X = 3.5$ using the selected split?

Problem 6. Confusion Matrix and Performance Measures

A classifier is evaluated on 300 observations, of which 60 belong to the positive class.

The classifier correctly identifies 45 positive observations and incorrectly classifies 30 negative observations as positive.

1. Construct the complete confusion matrix.
2. Calculate the accuracy, precision, recall (TPR), and false-positive rate (FPR).
3. Calculate the F_1-score using
$$F_1 = \frac{2\,\text{Precision}\,\text{Recall}}{\text{Precision} + \text{Recall}}.$$
4. Consider a classifier that always predicts the negative class. Calculate its accuracy and recall.
5. Based on these results, explain why accuracy alone can be misleading when classes are imbalanced.

Problem 7. ROC Curves and AUC

A classifier assigns the following scores to six observations:

Observation	True label	Score
A	1	0.90
B	0	0.80
C	1	0.70
D	0	0.60
E	1	0.40
F	0	0.20

The classifier predicts class 1 whenever
$$s(x) \geq t.$$

1. For thresholds
$$t = 0.85, \quad 0.65, \quad 0.45, \quad 0.30,$$
calculate the TPR and FPR.
2. Plot the corresponding ROC points, including the two endpoints $(0, 0)$ and $(1, 1)$.
3. Calculate the empirical AUC by comparing every positive-negative pair of observations.
4. Interpret the AUC as a probability.
5. What would happen to the ROC curve if every score were replaced by $2s(x) + 1$?

Problem 8. Logistic Regression and Decision Boundaries

Consider a logistic regression model with two features:

$$p(x_1, x_2) = \frac{1}{1 + \exp[-(x_1 + 2x_2 - 3)]}.$$

The model predicts class 1 whenever

$$p(x_1, x_2) \geq t.$$

1. Derive the decision boundary for $t = 0.5$.
2. Calculate the predicted probabilities for
$$A = (1, 1), \qquad B = (3, 1).$$
3. Find the value of t for which the decision boundary becomes
$$x_1 + 2x_2 = 4.$$
4. Determine the predictions for A and B under this new threshold.
5. Explain geometrically how changing the classification threshold affects the decision boundary.

Problem 9. Logistic Loss and Classification Error

Two classification models are evaluated on four observations.

They produce the following estimated probabilities of class 1:

Observation	True label	Model A	Model B
1	1	0.55	0.90
2	0	0.45	0.30
3	1	0.60	0.80
4	0	0.40	0.70

Both models use a classification threshold of 0.5.

The average logistic loss is

$$L = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right].$$

1. Calculate the classification error of both models.
2. Calculate the average logistic loss of each model.
3. Which model has the lower classification error? Which has the lower logistic loss?
4. Explain how it is possible for one model to have a lower classification error but a higher logistic loss.
5. Would changing the classification threshold affect the logistic loss computed from these probabilities? Explain.

Problem 10. Cross-Validation and Model Selection

A researcher considers three values of k for a k-NN classifier:

$$k \in \{1, 5, 15\}.$$

Using 4-fold cross-validation, the numbers of classification errors on the validation folds are:

Model	Fold 1	Fold 2	Fold 3	Fold 4
$k = 1$	4	8	5	7
$k = 5$	5	4	6	3
$k = 15$	7	5	8	6

Each validation fold contains 50 observations.

1. Calculate the average validation error rate for each value of k.
2. Which value should be selected by cross-validation?
3. Suppose the full dataset contains 250 observations, with 50 reserved for testing. How many observations are used to train the model within each cross-validation fold?
4. After selecting k, how many observations should be used to train the final model before test evaluation?
5. Suppose the selected model makes 7 errors on the test set. Calculate its test error rate. Explain why this test error should not be used to choose another value of k.